Dynamic Lexical Framework to Evaluate the Evolution of Emotions in Twitter

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ABSTRACT

Human emotions and sentiments are dynamic by nature. Nowadays, social networks have become a key resource for human communication and a faithful representation of this dynamism. This fact poses major challenges to those systems addressing sentiment analysis. Therefore, having systems capable of inferring this dynamism has become a key issue. In this paper we introduce Emoweb 2.0, a prototype for dynamic sentiment analysis of Twitter data. A well-known lexicon is taken as starting basis and new words are appended by an unsupervised learning algorithm governing the process. Sentiment values of new words are calculated and dynamically updated depending on the trends detected. Tweet sentiment scores are also computed during the process. A visualization module is included to observe word sentiment fluctuations over time. The experiment performed is based on the ongoing COVID-19 pandemic showing promising results.

CCS Concepts

• Information systems • Information retrieval • Retrieval tasks and goals • Sentiment analysis • Computing methodologies • Artificial intelligence • Knowledge representation and reasoning • Natural language processing • Machine learning • Learning paradigms • Unsupervised learning

Keywords

Sentiment analysis; Combination of information; Unsupervised learning system; Knowledge representation; Continuous dynamic system.

1. INTRODUCTION

Communication process represents an inherent essential need arising from human relationships. The process itself may involve several methods to express and effectively allow the phenomena to happen. The emotions, which come into existence as instinctive physical responses to external events, exemplify one of these methods [1]. The conscious linking process of emotions to thoughts also constitutes a major example of the foregoing, getting materialized by a verbalization which is referred to as sentiments [2].

The digital revolution and the unremitting technical development of the Internet resulted in new virtual communication systems at users' disposal which completely broadened human social interactions. Social networks represent the most prominent exemplification in this matter. Among the available platforms, Twitter stands out with an estimated 500 million messages being published per day [3]. In this regard, Twitter emerged as a valuable resource on which sentiment analysis techniques can be applied [4].

Human emotions and sentiments evolve over time and present a high dependency on external events. In Twitter, these facts are also observed when latest trends are being discussed and a high sensitiveness to events happening worldwide are shown in published tweets. Therefore, it becomes a key issue to develop an infrastructure completely aware of this inherent dynamism and truly capable of capturing the evolution in sentiments for a successful detection of trends.

For these purposes, in this paper a framework called EmoWeb 2.0 is introduced. The prototype is addressed to perform dynamic sentiment analysis of Twitter data. It takes its foundations on an unsupervised learning algorithm responsible for appending new words to an initial lexicon (SenticNet [5]). Sentiments calculated for the new words are dynamically updated depending on trends detected in Twitter data. The framework is also capable of providing sentiment scores to processed tweets according to the word sentiment values stored in the lexicon.

The architecture presented follows a modular approach to divide the process in several stages. The first module is responsible for gathering the Twitter data to be processed internally. The second module focus its actions on actually processing these data. The process involves the use of Natural Language Processing (NLP) techniques to obtain the relevant words to be registered in the lexicon. The third module is accountable for computing sentiment values for the detected words as well as for updating the internal archive for trends detection. A separate visualization module is also available to generate sentiment-time graphs of selected words

The ongoing coronavirus pandemic has been selected as the main subject to orchestrate the experiment performed. Being without a doubt the most important topic of 2020, the still remaining unanswered questions and the constant updates on this subject emphasise the need for a system strongly skilled to detect the evolution of sentiments in Twitter in this respect.

The rest of the paper is organized as follows. Section II provides a brief overview of the work related to the proposal. Section III describes the foundations of the developed framework by

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detailing its modules and components. Section IV presents an experiment on the selected domain to illustrate the viability of the framework. Section V concludes and indicates potential future research lines.

2. RELATED WORK

Sentiment Analysis constitutes a field of investigation under NLP with the aim to extract subjective information expressed in texts written by humans. Its duties also refer to the study of the influence that texts may exert on readers and their respective feelings. Multiple applications have benefited from the activities involved in this field. Some instances reflecting these contributions are brand monitoring and reputation management (which can keep track of opinions given by users to certain products and apply corrective measures where required), market and competition research (which leads to finding out the strategies and actions the competitors are taking) and social media monitoring, among many others.

The information retrieved from Sentiment Analysis techniques may be classified according to different criteria. One strategy can be focused on emotions detection. The emotions can be classified into anger, fear, joy, repulsion, sadness and surprise [1]. The intrinsic complexity on this categorization leads to following a simpler approach less prone to potential discrepancies. Therefore, representations based on the polarity of emotions (positive, neutral and negative) are normally used. This convention also facilitates the subsequent extrapolation of the sentiments detected into numeric scales [6].

With respect to the actual implementations of Sentiment Analysis, the most common approaches are based on dictionaries and Machine Learning (ML) techniques. Dictionaries are a collection of prestored words associated with their sentiment score or polarity. These dictionaries are usually referred as to lexicons. Well-known examples of general purpose lexicons are depicted by SentiWordNet [7], SenticNet [5], WordNet Affect [8], Opinion Lexicon [9], AFINN [10], SentiStrength [11] and SentiFul [12]. Sentiment techniques based on dictionaries compute the polarity of sentences by averaging the polarity of the words matching dictionary contents. The limited set of words stored in the lexicons represents one major shortcoming that could reduce the results accuracy when words are not found in it. Another downside is shown by the fact that a poor recognition of affect is obtained when linguistic rules are involved [13].

As for the ML approaches, statistical models are used to predict sentiment values. The model is initially trained with some preparation data which usually involves a collection of tagged corpora. The second step responds to the actual classification of words by the model. Some instances representing this approach are convolutional neural networks [14] and Bidirectional Encoder Representations from Transformers (BERT) [15]. It is worth noting that training shows to be a very important phase, specially when specific contexts are considered. In addition, it becomes important to reflect the polarity changes occurring over time in the training data to avoid drawbacks in this regard [16].

The combination of both strategies has been addressed by the scientific community in search for hybrid feasible solutions solving some of the individual aforementioned drawbacks [13]. In these approaches, the polarity of a given text gets inferred by firstly applying dictionary based techniques followed by ML strategies applied to those words not found in the lexicon.

The framework presented in this paper follows a similar hybrid approach to proceed with sentiment analysis. It presents an evolution of a prototype called EmoWeb [17]. EmoWeb was originally conceived to provide a visual framework for dynamic emotional web analysis based on sentiment evaluation of textual content extracted from websites. Taking as a starting basis a well-known lexicon (SentiWordNet [7]), the framework implemented text analysis processes followed by an unsupervised learning algorithm responsible for incorporating new words to the lexicon and calculating (or updating) their sentiment polarization and strength over time.

Two key contributions of EmoWeb were the ability to make the lexicon grow dynamically and the inclusion of time as a considered magnitude. The former, allowed the prototype to incorporate new words detected during sentiment analysis processes which resulted in future incremental improvements of the accuracy of its outcomes. The latter, added a significant value to be able to detect trends and incorporated the memory concept (sentiments calculated for the words detected on a particular calendar day would take into consideration the calculations of previous days).

The evolution of EmoWeb presented in this paper (called EmoWeb 2.0) preserves the original concept (including the unsupervised learning algorithm) and performs all necessary adaptations to let it work with data coming from Twitter. The hybrid approach gets manifested by firstly taking SenticNet [5] as initial lexicon to then calculate the sentiment values of the new words according to the sentiment scores of the tweets where they were detected. This feature reminds to the statistics approaches, where the polarity of a word is computed according to the frequency of words appearing close to it (cooccurrence). This results in the calculation of context-based sentiment values for the new words detected.

As a consequence, EmoWeb 2.0 is capable of learning new words and their respective sentiment values. In addition, the framework also provides support to the dynamic nature of sentiments by updating the polarity values stored in the lexicon depending on the trends detected in input texts. This circumstance translates into having a dynamic lexical framework able to adapt its knowledge over time.

3. PROPOSED ARCHITECTURE

The proposed architecture for EmoWeb 2.0 is illustrated in Fig. 1. It is worth noticing the use of SenticNet as a general purpose lexicon. In addition, the English language has been taken as the basis for the lexicon construction and all the Twitter data retrieved.

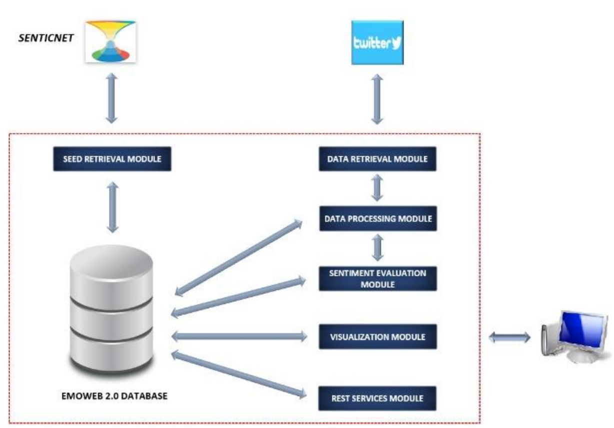

Figure 1. EmoWeb 2.0 Architecture.

In a nutshell, the overall process itself can be summarized by stating that relevant words are gathered from the incoming sets of tweets and then stored in the lexicon (if not there before) along with a computed sentiment value. The sentiment is estimated from the former word sentiment stored in the lexicon (if exists) and the sentiment scores of the tweets where the word appeared. Words are considered as relevant depending on the Part of Speech (PoS) label inferred from the NLP process. These techniques likewise include lemmatization activities to procure corresponding lemmas. For its part, sentiments associated with words are represented by numerical values in $[-1,1]$. The lexical framework is conceived to be ideally exposed to sets of tweets belonging to consecutive calendar days, one at a time. This feature offers solid background to both compare the words detected each day and provide unique proficiency to run calculations related to the evolution of sentiments over time. This evolution relies upon whether each word itself is being detected in Twitter data pertaining to different calendar days. In this context, a penalization factor is applied to the sentiments associated with lexicon words not being identified in subsequent tweet sets (i.e., the words are exerting less influence in the community due to lack of use). On the contrary, the sentiments associated with the lexicon words detected in subsequent tweet sets are recalculated by considering a compensation factor. In addition, it is worth noting that the framework calculates the sentiment of input tweets as the average of the sentiments of their constituent words.

In light of the foregoing, the dynamic nature of the lexical framework therefore resides in the ability to make the lexicon grow by adding new words resulting from Twitter data analysis. Moreover, the concept is reinforced by the inherent dynamism arising from the evolution of emotions associated with words. These features provide the framework with the capability to detect trends and qualifies it to generate sentiment-time graphs per word.

Delving into the internal components of EmoWeb 2.0, the entire process has been divided into independent stages executed in a sequential manner. Each internal module is responsible for the tasks solely falling into one of the stages. This leads to allowing the architecture to likewise serve as a clear indicative of the internal process followed. The database acts not only as a storage resource for generated data but also as a critical passive coordination and cohesive member of the system, specially when the modules require access to the corresponding preceding stage results.

The Seed Retrieval Module is responsible for creating the initial lexicon by persisting SenticNet knowledge-base into the database. The word entries are stored along with their polarity (values in $[-1,1]$). As for the Data Retrieval Module, it is in charge of gathering the relevant data to be processed internally. As a core design decision strategy, each input set will correspond to tweets created on the same calendar day. This fact allows the system to effectively detect trends.

With regards to the Data Processing Module, its responsibility lies on processing each tweet set presented to the system. Fig. 2 illustrates the internal process followed to cope with each tweet.

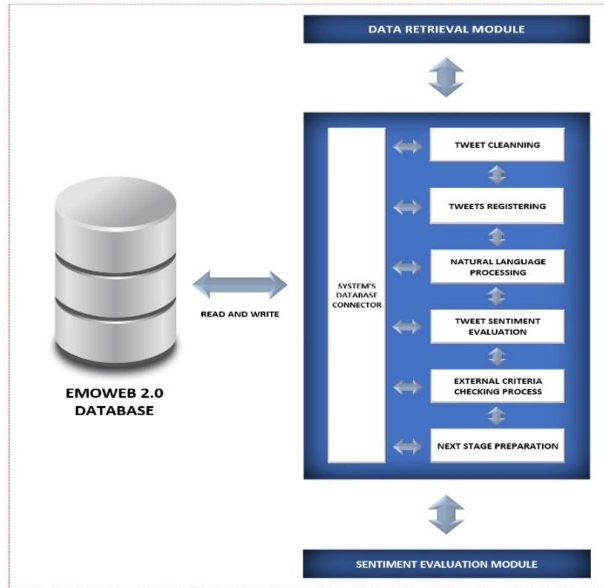

Figure 2. Data Processing Module Architecture.

It starts by firstly applying basic preprocessing to the content of the tweets. These activities include the cleaning of not relevant information (smileys, emojis, special characters, mentions, hashtags, etc.). The resulting cleaned text along with some other details (external ID, tweet date, country, etc.) are then stored in the database. Subsequently, the NLP process is triggered. Actions under NLP include the morphological analysis of the input texts by means of lemmatization and PoS tagging methods. It also covers the elimination of stopwords detected. Resulting words from the NLP process are then looked up in the lexicon and registered if not present before (new words are given an initial sentiment equal to 0). Finally, tweet sentiment values are calculated.

Assuming that the input set of tweets belongs to calendar day d, then tweet sentiments are calculated as follows:

$$s_i^{(d)} = \frac{1}{N}\sum_{j=1}^{N} s_{w_j}{}^{(d)} \quad (1)$$

Where $s_i^{(d)}$ represents the sentiment value of the input tweet i belonging to calendar day d, N refers to the number of words detected after the NLP process of it and, finally, $s_{w_j}{}^{(d)}$ details the sentiment value stored in the lexicon for word w_j on the calendar day d. It is worth noting that if the same tweet is processed on a different day, the sentiment value obtained may vary as word sentiment values can be different.

Focusing on the Sentiment Evaluation Module, its actions takes place at the end of the day after processing the complete input tweet set and having registered all the new words. Its duties are addressed to update the sentiment values of the lexicon words with all the knowledge gathered during the day and also to archive it accordingly.

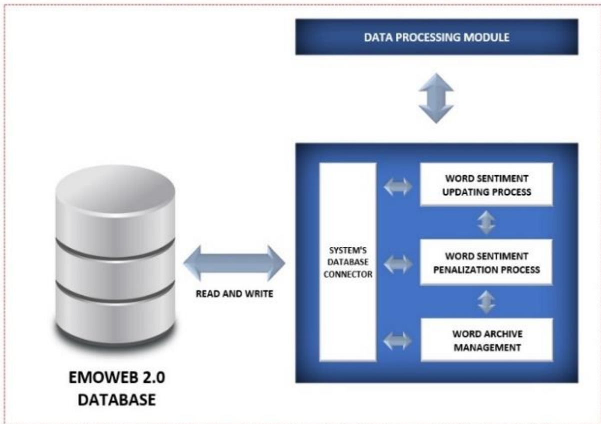

Figure 3. Sentiment Evaluation Module Architecture.

The sentiment values of lexicon words are updated as follows:

$$s_w^{(d+1)} = (1-\alpha) \cdot s_w^{(d)} + \alpha \cdot \sum_{j=1}^{N} s_j^{(d)} \qquad (2)$$

where $s_w^{(d+1)}$ details the sentiment value of word w to be used during the following calendar day processing d + 1, $s_w^{(d)}$ describes the current stored sentiment for word w, N represents the number of input tweets created on calendar day d where the word w was detected, $s_j^{(d)}$ refers to the sentiment value calculated for tweet j on calendar day d and, finally, $\alpha \in [0,1]$ represents an internal parameter to adjust the weight and impact of the terms in the equation. It is relevant to note that the parameter α allows modulation of the framework memory by balancing between the former knowledge and the new knowledge (taken as a the sum of tweet sentiments where the word appeared).

After completion of the sentiments updating process, management tasks related to the internal archive are triggered. These activities basically consist in copying the updated lexicon into the archive along with the calendar day on which the set of input tweets were created. As a result, this serve to detect trends and consolidate a strong support for the time-based analysis.

Finally, with respect to the Visualization Module, it offers all necessary tools to generate sentiment-time graphs for each word stored in the archive.

4. EXPERIMENTS

The experiment pursued the achievement of two separate goals. The first objective resided in evaluating the performance of the framework by comparing the tweet sentiment scores calculated to expert opinions. The second goal was addressed to show the dynamism of the lexical framework by representing the sentiment evolution over time of some selected words of interest.

The exercise exposed the framework to tweets dealing with a specific subject. In this respect, the coronavirus pandemic and more accurately, the tweets created during its second wave were selected [18]. Datasets were downloaded from an existing running project from the IEEE. Tweets were collected by monitoring keywords and hashtags related to the pandemic which ensured proper adequacy. The IEEE published one dataset per day which includes tweet IDs and sentiment scores computed by TextBlob (values in [−1,1]). These facts revealed perfect consistency with the proposed daily process scheme and the internal sentiment score calculated for tweets (same scale allowed comparison). Nonetheless, tweet datasets required hydration before processing.

For our purposes, a training and a test dataset were prepared. The training set got defined by the 30 datasets published by the IEEE during November 2020. The total amount of tweets processed for this aim amounted to 140,429. As for the test, the 5 datasets ranging from December 1st to December 4th were utilized, configuring a total amount of 19,043 tweets processed. As for the internal parameter α, it is set to 0.5 to equally balance the previous and the new knowledge available.

In order to accomplish the first goal, the experiment assessed the framework performance by following two different strategies. The first approach consisted in skipping the training phase and just processing the test dataset corresponding to December. Therefore, in this scenario tweet sentiment scores were calculated on the basis of the initial knowledge provided by the SenticNet lexicon and the words learnt during the process itself. The second strategy, on the contrary, considered a prior framework training with November datasets before actually processing the test dataset. Tweet sentiment score results obtained by the both strategies were then compared to the expert opinions available (TextBlob values) and assessment metrics estimated. The Table I illustrates the results obtained. Some promising expectations can be derived for the foreseeable future as, for instance, computed accuracy reflected a 5% increase when the framework was trained in advance. This initial results seem to underline the fact that predictions get improved when a better starting basis is available.

Table I. Metrics for December test results

	No prior training	Prior training
Accuracy	0.4303	0.4512
Precision	0.4138	0.4379
Recall	0.4220	0.4371
F1-score	0.4178	0.4375

As regards to the Table II, it shows the number of words stored in the initial lexicon and the way this number was incremented when the datasets were processed. From an initial SenticNet seed of 100,000 words, the process of the training dataset (corresponding to November) increased the knowledge-base by 35,885 words. In respect to the test dataset process (related to December), it brought 3,667 new words to the lexicon.

Table II. Words stored in internal lexicon

Initial lexicon (SenticNet)	100,000
After training with November	135,885
After testing with December	139,552

In reference to the second goal, the Fig. 4 portrays the sentiment evolution of some coronavirus related words obtained after processing the training and the test datasets. In conjunction with the results presented in Table II, the dynamism of the lexical framework is proven.

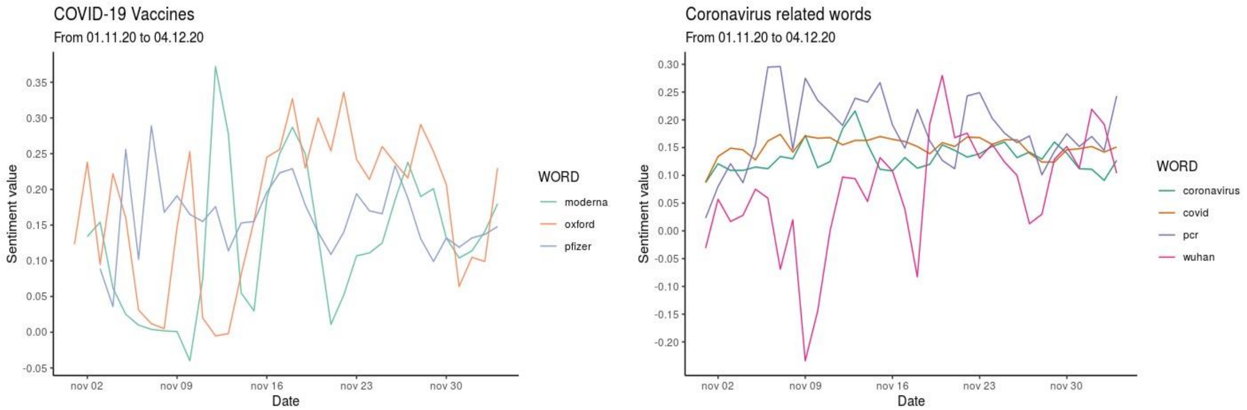

Figure 4. Sentiment values for some COVID-19 and coronavirus related words.

The Fig. 4 depicts that two different set of words were selected. The first group considers the current most advanced vaccine candidates for the COVID-19 (i.e., Moderna, Oxford and Pfizer). Regarding the second group, it includes some relevant words in this context. Focusing on the first group, it is relevant to note that none of the vaccines result in a negative sentiment during the period under analysis. The Pfizer vaccine shows a more stable sentiment adhered to it which can be interpreted in terms of more trust or less uncertainty. From the very beginning, Pfizer vaccine was claimed to reach an alleged 95% efficacy. The others, conversely, were distant from that figure. These facts could lead to having more fluctuations on the Moderna and the Oxford vaccine associated sentiments.

As concerns the second group of words, PCR exhibits sentiment variations potentially derived from the false negatives reported. As respects to Covid and Coronavirus, they present a relative constant sentiment which could be explained by population acceptance of current situation and envisaged hope drawn by applicable solutions available. On a final note, Wuhan gets associated with a negative perception as the presumed pandemic provenance.

Trends detected disclose a promising baseline for future exercises. In the first place, the examination of longer periods of time may reveal further features. In the same manner, a proper knowledge of the major events occurring during the period of time under study may serve as immediate contextualization and explanation of the sentiment fluctuations detected. Finally, cross studies of related word trends may bring to light additional information and eventually confirm correct framework functioning.

5. CONCLUSIONS

This paper has introduced a framework to provide dynamic emotional analysis based on the sentiment evaluation of Twitter data. An unsupervised learning algorithm has been implemented to govern the process by incorporating new words to its lexicon. Stored words get associated with a sentiment score which evolves over time depending on the trends detected. Tweets processed also receive a sentiment score which can be compared to expert opinions for assessment purposes.

A basic experiment focused on an specific subject has been performed to validate the overall design concept and the main goals pursued by the proposal. The results gathered reveal a promising performance and a good correlation with expected behaviour. The framework presented has shown its capability to dynamically increment its knowledge base by registering new words in its lexicon. Its aptitudes to provide sentiment scores to tweets and stored words have also been proven and assessed by comparison with expert opinions. The detection of trends likewise strengthens the validity of the proposal by producing customized visualizations.

All the previously mentioned establishes solid grounds on which future further developments may be built upon to upgrade the current functional prototype. As a preliminary point, an initial context specific lexicon could be implemented. As an extension of the aforementioned, having a pool of expert opinions would also provide a richer reference to which compare EmoWeb 2.0 results. In this respect, integration processes with BERT [15] and µTC [19] frameworks are currently being developed to increase the number of opinions available. With regards to trends, longer periods of time could be considered to retrieve more accurate observations of sentiment fluctuations.

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